

Roll No.

TAE-714

**B. TECH. (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
VEHICLE BODY ENGINEERING AND SAFETY**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks

1. (a) List down the various aluminium alloys and explain their automotive applications. (CO1)

OR

- (b) How does silicon affect the properties of aluminium ? Explain few improved properties. (CO1)

2. (a) Explain the working of extrusion process with a suitable diagram. (CO1)

OR

- (b) Why are composites so famous for automotive applications ? Also, classify various automotive composites. (CO1)

3. (a) Explain the importance of ergonomics in vehicle design. (CO2)

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(2)

OR

(b) What are the various domains wherein ergonomics is essentially required ? (CO2)

4. (a) Explain vehicle packaging with suitable examples. (CO2)

OR

(b) Explain "Blind Spots" of a car with suitable diagram. (CO2)

5. (a) How you will perform visibility test for driver ? (CO2, CO3)

OR

(b) What are the various aerodynamic forces that act on an automobile at high vehicle speed ? (CO2, CO3)